

Sound Waves

Make a Wave Machine!



What is Sound?

Sound is a vibration that travels
through air, water, or solids
to reach our ears!



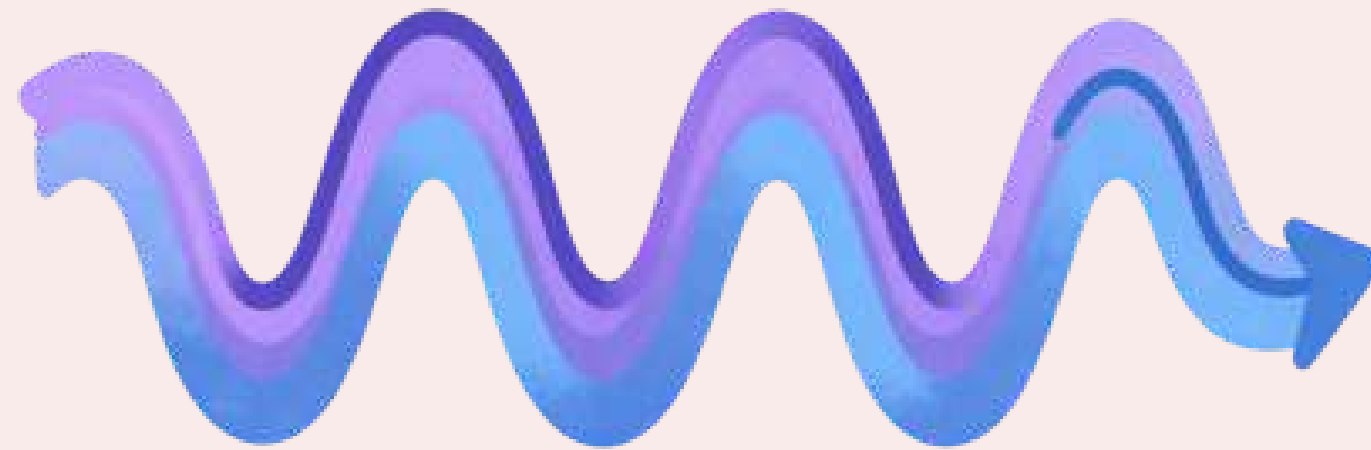
How Does Sound Travel?

Sound moves in WAVES!
Just like ripples in a pond
when you throw a stone.



What Are Waves?

Waves are patterns of movement
that carry energy from
one place to another.

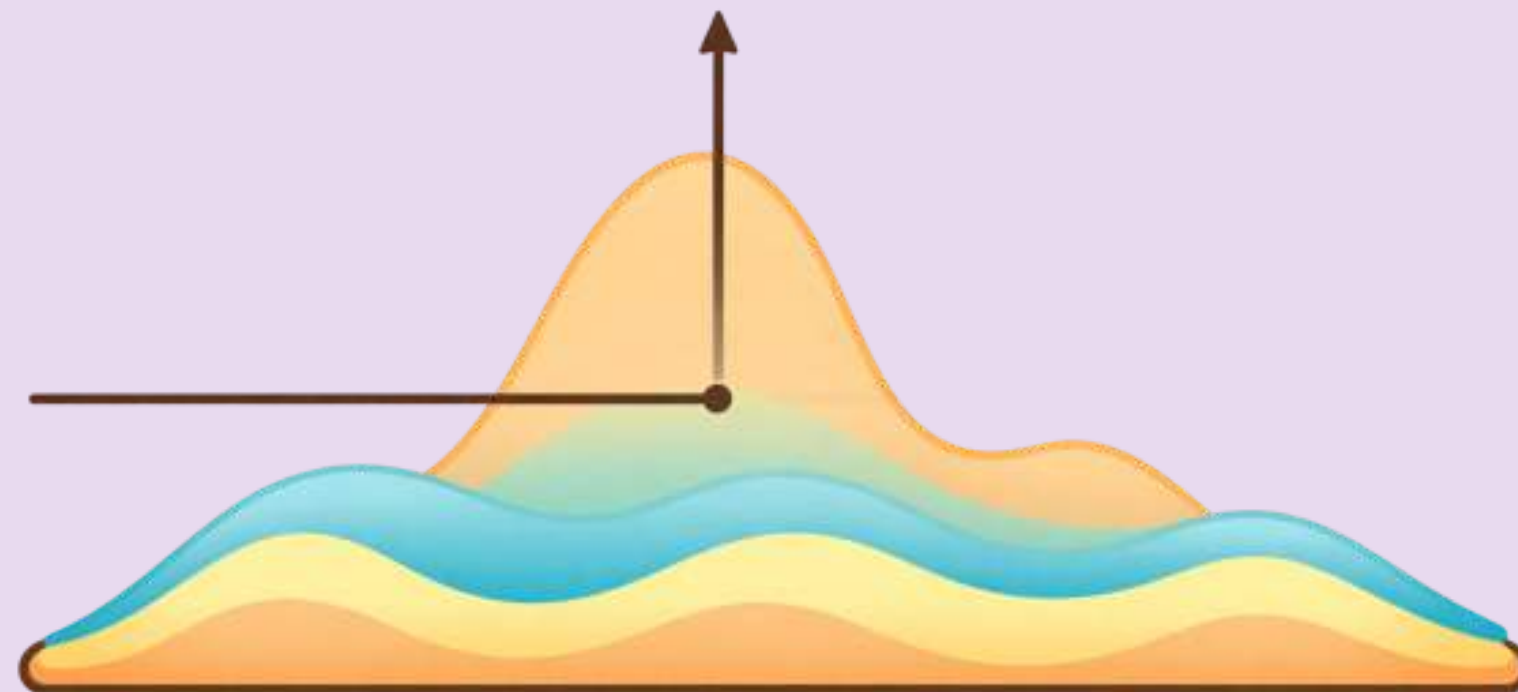


Parts of a Wave

Crest = the top of a wave

Trough = the bottom of a wave

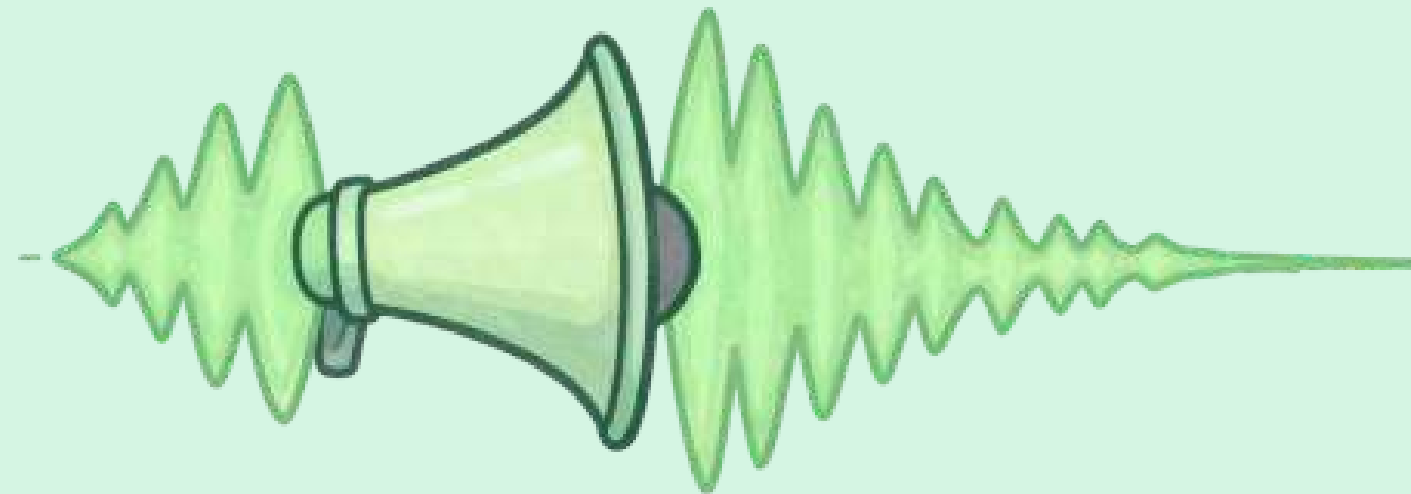
Wavelength = distance between crests



Loud vs Soft Sounds

Loud sounds = BIG waves (high amplitude)

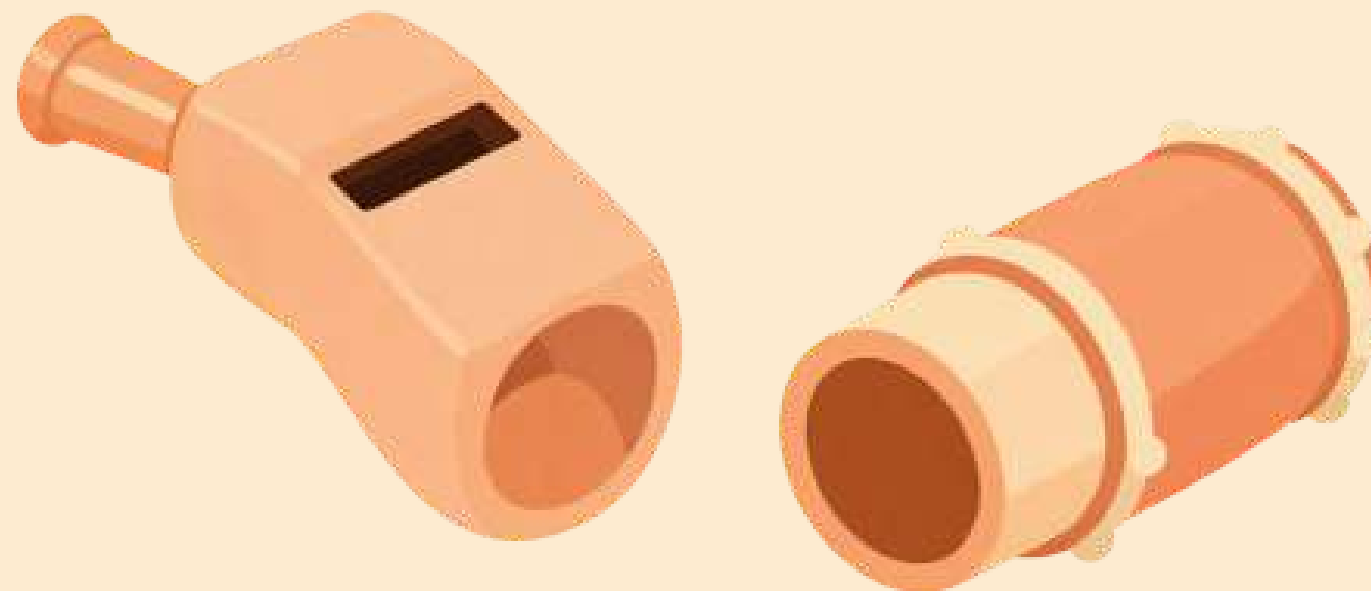
Soft sounds = small waves (low amplitude)



High vs Low Pitch

High pitch = waves close together (like a whistle)

Low pitch = waves far apart (like a drum)



Let's Build a Wave Machine!

We will make a simple wave machine
using everyday materials to SEE
how sound waves move!



Materials You Need

30 wooden skewers or sticks

A long strip of duct tape (1 metre)

Small candy or clay balls

A table or flat surface



Step 1: Lay the Tape

Lay a long strip of duct tape
sticky side UP on the table.
Keep it straight and smooth!



Step 2: Place the Skewers

Place skewers across the tape
about 2 cm apart from each other.
Press them down gently.



Step 3: Add the Balls

Stick small clay balls or candy on BOTH ends of each skewer. These are your wave markers!



Step 4: Seal It!

Place another strip of tape
ON TOP to sandwich the skewers.
Now pick it up carefully!



Step 5: *Make Waves!*

Hold one end steady.
Tap a skewer on the other end.
Watch the wave travel across!



What Did You Observe?

The wave moves from one end to the other!

Energy transfers through each skewer
just like sound through air!



Great Job, Scientists!

You built a wave machine!
Now you know how sound
travels as waves of energy!

